

Cecilia Maltese Invrea

01/12/2002 • Italian Nationality
Munich, Germany • +39 331 7378891 • cecilia.malteseinvrea@gmail.com

EDUCATION

2025–2027	TUM – Technische Universität München <i>Master's degree in Automotive Engineering</i>	Munich, DE
2021–2025 Feb	Polimi – Milan Polytechnic University <i>Bachelor in Automation Engineering, Major in Robotics</i>	Milan, ITALY
2024 Oct–Nov	ESA-ESTEC Academy <i>Robotics Workshop – Design and operation of an ExoMy rover; AI model training</i>	Noordwijk, NL
2024 Sept–Mar	ETSI – Escuela Técnica Superior de Ingeniería <i>Erasmus+ Exchange Program – Applied Electronics for Automation</i>	Seville, SPAIN
2019 Jul–Dec	Sunshine Beach High School <i>Exchange Program</i>	Noosa, AUS
2011–12	La Scuola d'Italia G. Marconi <i>Exchange Year</i>	New York, USA

WORK EXPERIENCE

2026 Jan–present	Entrix GmbH <i>Working Student in Tech and Operations</i> Configure and commission hardware for BESS (Battery Energy Storage System) customer installations. Coordinate with customers through technical onboarding to ensure correct data acquisition setup. Validate incoming energy data streams: network checks and plausibility analysis. Design and develop internal web infrastructure (front-end and back-end) to streamline and automate day-to-day operational workflows.	Munich, DE
2025 Jan–Jun	OHB Italia <i>Intern in Mechatronics Department</i> Mechatronic development of a lunar regolith excavation, manipulation and transport system for future missions. Defined system architecture; verified in simulation; integrated test setups and executed functional checks. Multibody simulations of a linear positioning system using Hexagon's ADAMS; aided in manufacturing phase.	Milan, ITALY
2024 Mar–Sept	ESA European Space Agency – ESRIN <i>Intern, Ground Segment & Mission Management – Directorate of Earth Observation Programmes</i> Worked on the DestinE project leveraging AI, LLMs and satellite data. Developed a full-stack web application (HTML/CSS/JavaScript; Python/Flask); implemented Knowledge Graphs in Python; built an interactive satellite data map. Research topics: Villa Adriana, Sirente Crater, Mt. Vesuvius.	Frascati, ITALY

ADDITIONAL EXPERIENCE

2017	UN Model , CW Mun Emirates, Associazione Diplomatici	Abu Dhabi, UAE
2016	N.P.H. and Fondazione Rava , Volunteering Project Certificate	Dominican Rep
2015	Quantum Learning Super Camp , Program and Certificate	Rome, ITALY
2014	Volunteering for Peace Vietnam , Volunteer at Pagoda Orphanage	Saigon, VIETNAM
2009	Round the World Trip : 80 days, 5 Continents	

SKILLS

IT	Programming: C, Java, Python, HTML, CSS, JavaScript, MATLAB and Simulink Softwares: ADAMS, LTspice, ROS2, Gazebo, Webots, Arduino IDE, CodeSys, Bash, Git, Docker, VSCode, Excel OS: macOS, Windows
Languages	Italian (Native) English (C1 – IELTS 8.5/9) Spanish (Fluent) French (B1 – DELF) German (A2)

INTERESTS

Travel & Cultural Exploration (30+ countries) | Reading | Sports | Astronomy | Foreign Languages

ENGINEERING PROJECTS

2026 Mar–Aug	Dodo Alive! Teaching a Dodo to Walk (and Backflip) Reinforcement-learning locomotion for an 8-DOF bipedal robot in NVIDIA IsaacLab. 256 parallel robots trained in simulation with no hand-crafted gaits, reward-driven forward locomotion, targeting sim-to-real transfer.	TUM, Munich
2026 Mar–Aug	Biologically Inspired Learning for Humanoid Robots Reimplementation of the RL-DT paper (Reinforcement Learning with Decision Trees): a sample-efficient, model-based method that generalizes action effects across states via decision trees, with targeted exploration for fast learning. Deployed on an Alnex humanoid in a penalty-kick goal-scoring scenario.	TUM, Munich
2026 Mar–Aug	Acoustic Click-Detection for Industrial QA Deep-learning classifier for microphone recordings robust to industrial noise. Full ML pipeline: data exploration, preprocessing, feature engineering, training, and noise-scenario evaluation. Conventional mic plus contact microphone on the gripper; explored robot joint-position and force-torque streams as additional modalities.	TUM, Munich
2026 Oct–Feb	Robot Programming and Control for Human Interaction Robotic manipulation and human-robot interaction from theoretical foundations through simulation to hands-on validation on the KUKA-DLR light-weight robot. Topics: robot modelling and parameter identification, position/torque control, Cartesian Impedance Control, collision detection, reactive path generation, state machines for task programming.	TUM, Munich
2024 Oct–Nov	Design and Operation of an ExoMy Rover 3D-printed rover modelled after the Rosalind Franklin Mars rover. Developed movement algorithms and applied Machine Learning for object recognition on Martian terrain. Selected among 300+ European students.	ESA ESTEC, Noordwijk
2023 Sept–Dec	Sensor-based Automation & Energy-efficient Optimization – RC Vehicle Collaborative project automating small racing cars via microprocessors and sensor integration (ultrasound and distance sensors), with focus on energy efficiency enhancements. Responsible for the sensors unit.	ETSI, Seville
2022	Globe at Night Project International citizen-science campaign to raise public awareness of light pollution impact.	Milan, Italy
2022	Signal Analysis and Transfer Function Modelling – MATLAB Comprehensive signal analysis on diverse input data and development of transfer function models. Data processing, system identification, and graphical visualisation of modelled system behaviour.	Polimi, Milan
2021	Roguelike Videogame Development Conceptualisation and development of a roguelike videogame in C#.	Polimi, Milan